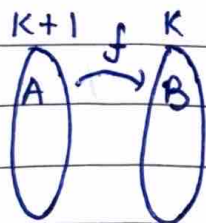


# Unit - 2

\_\_/\_\_/

- Pigeon hole principle  
Let  $K \in \mathbb{Z}^+$ . Assume  $K+1$  objects are to be placed in  $K$  boxes then there is at least one box containing two or more objects.
- Proof by contradiction  
Let all boxes have one object  
Total no of objects  $\rightarrow K$  which is a contradiction  
 $\rightarrow K+1$  objects  
 $\therefore$  At least one box has more than one object

Q A func  $f$  from a set with  $K+1$  or more elements isn't one-to-one



Acc to pigeon hole principle

no of objects =  $K+1$

no of boxes =  $K$

At least 2 elements of set A should be assigned to some element of set B.

Q Show that among any group of 367 people there must be at least 2 with same birthday

By PHP

no of objects (people)  $\rightarrow K+1$  (367)

no of boxes (days)  $\rightarrow K$  (366/365)

Hence at least 2 have same birthday

- Extended / Generalised pigeon hole principle  
If  $N$  objects are placed into  $K$  boxes then there is at least one box containing  $\left(\frac{n-1}{K} + 1\right)$  objects or  $\left\lceil \frac{N}{K} \right\rceil$  objects.

GIF

Q Among 50 people what's the no. of people that must be born on same month

$$\left[ \frac{50}{12} \right] = [4.16] = 5$$

Q How many students in a class must there be to ensure that 3 students get same grade A, B, C, D & F

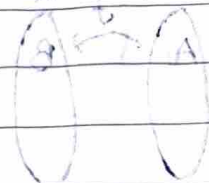
$$\left( \frac{N-1}{K} + 1 \right) = 3$$

$$N-1+5 = 3$$

5

$$1 + N + 4 = 15$$

$$N = 10$$



Q A box has 10R & 10Y balls. How many balls must be selected to ensure 3 balls of same colour.

$$\frac{N-1}{2} + 1 = 3$$

$$\Rightarrow N = 5$$

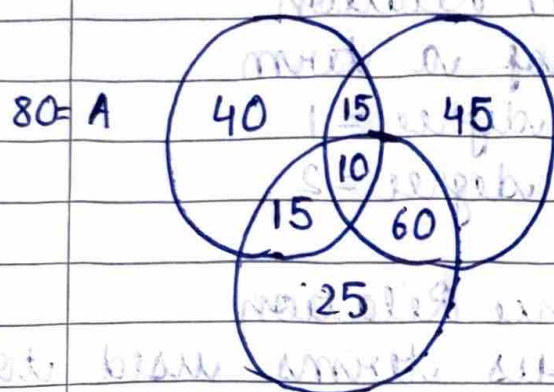
Q Min no of students each of which come from one of the 50 states must be enrolled in a university to guarantee that atleast 100 come from same state.

$$\frac{N-1}{50} + 1 = 100$$

50



Q In a group of 200 people each one is atleast accountant or sales manager or consultant. It's found 80 accountants, 130 sales managers, 10 are accountants & sales managers & consultants, 25 are acc & sales manager & 70 are manager and consultants and 110 are consultants. Find people who are accountants & consult.



$M = 130$

$$200 = 80 + 110 + 130 - 25 - 70$$

$$200 = 320 - 95 - x$$

$$x = 320 - 200 - 95$$

$$x = 120 - 95 = 25$$

25 are acc & consultants

Q No not divisible from 5, 7, 9 from first 500 nos

$$A \rightarrow \div 5$$

$$B \rightarrow \div 7$$

$$C \rightarrow \div 9$$

$$n(A) = 100 \quad n(A \cap B) = 14$$

$$n(B) = 71$$

$$n(B \cap C) = 7$$

$$n(C) = 55$$

$$n(A \cap C) = 11$$

$$n(A \cap B \cap C) = 1$$

$$n(A \cup B \cup C) = 100 + 71 + 55 - 14 - 7 - 11 + 1$$

$$= 100 + 126 - 31$$

$$= 195 \rightarrow \div 5, 7, 9$$

$$\text{Not } \div 5, 7, 9 = 500 - 195 = 305$$

## → Recurrence Relations

A relation which defines  $n^{\text{th}}$  term using one or more previous terms

eg  $\rightarrow a_n = a_{n-1} + a_{n-2}$

we have to give initial condition

eg  $a_0 = 1 = a_1$  or  $a_n = 3^n$

## → Degree of Recurrence Relation

It's highest power of a term

$a_n = a_{n-1}$  degree = 1

$a_n^2 + a_{n-1} = 0$  degree = 2

## → Order of Recurrence Relation

Number of previous terms used to define  $a_n$

$a_n = a_{n-1}$  order  $\rightarrow 1$

$a_n = a_{n-1} + a_{n-3}$  order  $\rightarrow 3$

## → Homogeneous & Non homogeneous equation

Homogeneous uses only previous terms equalling zero

eg  $\rightarrow a_n = c_1 a_{n-1} + c_2 a_{n-2} = 0$

or  $a_n - c_1 a_{n-1} - c_2 a_{n-2} = 0$

$a_n - c_1 a_{n-1} - c_2 a_{n-2} = 0 \rightarrow \text{Homogeneous}$

Non homogeneous equations uses an additional function

eg  $\rightarrow a_n - c_1 a_{n-1} - c_2 a_{n-2} = 2^n$

↓  
additional function



# → Linear Homogeneous Recurrence Relations

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = 0$$

$c_n \rightarrow$  constants

1. Assume  $a_n = r^n$

$$\Rightarrow a_{n-1} = r^{n-1}$$

$$a_{n-2} = r^{n-2}$$

2. Put these values in equation

3. divide both sides by  $r^{n-k}$

4. Find roots  $r_1, r_2$

Characteristic roots

$$\Rightarrow c_0 r^n + c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k} = 0$$

$\div$  BS by  $r^{n-k} = 1$

$$c_0 r^k + c_1 r^{k-1} + \dots + c_k r^0 = 0$$

$$\Rightarrow c_0 r^k + c_1 r^{k-1} + c_2 r^{k-2} + \dots + c_k = 0$$

distinct

Case I Real & ~~equal~~ roots  $(r_1, r_2) \rightarrow 2$  roots

Then

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

$$\underbrace{\hspace{10em}}$$

$$\hookrightarrow r_1 \neq r_2$$

$\alpha_1, \alpha_2$  constant

Case II Real & equal roots  $(r_1 = r_2)$

$$a_n = (\alpha_1 + \alpha_2 \cdot n) r^n$$

$$\underbrace{\hspace{10em}}$$

If 3 roots  $r_1 = r_2 = r_3$

$$a_n = (\alpha_1 + \alpha_2 + \alpha_3 n^2) r^n$$

$$\underbrace{\hspace{10em}}$$

case 4 3 roots  $\rightarrow$  2 equal & 1 distinct ( $r_1 = r_2 \neq r_3$ )

$$\underline{a_n = (\alpha_1 + n\alpha_2)r_1^n + \alpha_3 r_3^n}$$

case 3 Imaginary roots

$$\underline{a_n = \alpha_1 (A + iB)^n + \alpha_2 (A - iB)^n}$$

Q  $2a_n - 5a_{n-1} + 2a_{n-2} = 0$   $a_0 = 0$  &  $a_1 = 1$

$$2r^n - 5r^{n-1} + 2r^{n-2} = 0$$

$$2r^n - 5r^{n-1} + 2r^{n-2} = 0$$

$$\div r^{n-2}$$

$$2r^2 - 5r + 2 = 0$$

$$(r-2)(r-1) = 0$$

$$a_n = \alpha_1 (2)^n + \alpha_2 \left(\frac{1}{2}\right)^n$$

$$\Rightarrow 0 = \alpha_1 + \alpha_2 \Rightarrow \alpha_1 = -\alpha_2$$

and

$$1 = 2\alpha_1 - \frac{\alpha_1}{2} = \frac{3\alpha_1}{2} \Rightarrow \alpha_1 = \frac{2}{3}$$

$$\alpha_2 = -\frac{2}{3}$$

$$\Rightarrow a_n = \frac{2}{3} (2)^n - \frac{2}{3} (2)^{-n}$$

$$= \frac{2}{3} [2^n - 2^{-n}]$$

Q  $a_n - 7a_{n-1} + 10a_{n-2} = 0$   $a_0 = 0$   $a_1 = 3$

$$r^2 - 7r + 10 = 0$$

$$(r-2)(r-5) = 0$$

$$a_n = \alpha_1 2^n + \alpha_2 5^n$$

$$\Rightarrow 0 = \alpha_1 + \alpha_2 \Rightarrow \alpha_1 = -\alpha_2$$

$$3 = 2\alpha_1 + 5\alpha_2 = -3\alpha_1 \Rightarrow \alpha_1 = -1 \text{ & } \alpha_2 = 1$$

$$\Rightarrow a_n = 5^n - 2^n$$



$$Q \quad a_n - 4a_{n-1} + 4a_{n-2} = 0 \quad a_0 = 1, a_1 = 6$$

$$r^2 - 4r + 4 = 0$$

$$(r-2)^2 = 0 \Rightarrow r = 2, 2$$

$$r = 2, 2$$

$$a_n = (\alpha_1 + n\alpha_2) 2^n$$

$$1 = (\alpha_1) 2^0 \quad 6 = (1/2 + 2\alpha_2) 2$$

$$\alpha_1 = 1$$

$$\alpha_2 = \frac{5}{4}$$

$$a_n = \left(1 + \frac{5n}{4}\right) 2^n$$

$$Q \quad a_n = a_{n-1} + a_{n-2} \quad a_0 = 0, a_1 = 1$$

$$r^2 - r - 1 = 0$$

$$r = \frac{1 \pm \sqrt{5}}{2}$$

$$a_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$a_0 = 0 = \alpha_1 + \alpha_2 \quad \alpha_1 = -\alpha_2$$

$$a_1 = 1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) - \alpha_1 \left(\frac{1-\sqrt{5}}{2}\right)$$

$$1 = \alpha_1 \frac{2\sqrt{5}}{2}$$

$$\Rightarrow \alpha_1 = \frac{1}{\sqrt{5}}$$

$$\alpha_2 = -\frac{1}{\sqrt{5}}$$

$$a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n$$

Multiplicity  
 Jab poly. ke factor hre to us root wale factor ki  
 jo power max hogi wo us root ki multiplicity  
 hai eg  $\rightarrow P(x) = (x-3)^2 \rightarrow \text{multiplicity}_3 = 2$

$\rightarrow$  Particular solution for non homogeneous equations

eg  $\rightarrow a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$

1. If RHS is polynomial with degree 2 then

$a_n = P_1 n^2 + P_2 n + P_3$

2. If RHS is  $n^m + n^{m-1} + n^{m-2} + \dots$

$a_n = P_1 n^m + P_2 n^{m-1} + P_3 n^{m-2} + \dots + P_1 (n-1)^m + P_2 (n-2)^m + \dots$

eg:  $\rightarrow$  If RHS is polynomial in  $n$  with degree 2  
 $m = 2$

$a_n = P_1 n^2 + P_2 n + P_3 + P_1 (n-1)^2 + P_2 (n-2)^2 + \dots$

3. If RHS is  $(F_1 n^t + F_2 n^{t-2} + \dots) \cdot \beta^n$

$a_n = n^m (P_1 n^t + P_2 n^{t-1} + P_3 n^{t-2} + \dots) \beta^n$

where  $m$  is multiplicity of root  $\beta$  where  $\beta$  is characteristic root and it is the degree of polynomial on RHS

Q  $a_n - 2a_{n-1} = 3 \times 2^n \rightarrow \beta$  degree of  $3 \times 2^n = 0$   
 $x^n - 2x^{n-1} = 0$   
 $x^2 - 2x = 0$  if root  $x \neq \beta$   
 $(x-2)^1 = 0 \Rightarrow m = 0$   
 $x = 2 \rightarrow$  characteristic root  
 $\Rightarrow m = 1$

$a_n = n (P_1 n) 2^n = n (P_1) 2^n$

$a_{n-1} = (n-1) P_1 (2^{n-1})$

Now  $a_n - 2a_{n-1} = 3 \cdot 2^n$

$P_1 n 2^n - 2P_1 (n-1) 2^{n-1} = 3 \cdot 2^n$



$$P_1(n \cdot 2^n - (n-1)2^n) = 3 \times 2^n$$

$$P_1 = 3$$

$$\Rightarrow a_n = n \times 3 \times 2^n$$

Q  $a_n + a_{n-1} = 3n 2^n \rightarrow \beta = 2$

Put  $a_n + a_{n-1} = 0$  degree  $3n 2^n = 1$

$$a_n = r^n$$

$$r^n + r^{n-1} = 0$$

$$\div r^{n-1}$$

$$r + 1 = 0$$

$$r = -1$$

$$\beta \neq r$$

$$\Rightarrow m = 0$$

$$\Rightarrow a_n = (P_1 n^0 + P_2) 2^n$$

$$a_{n-1} = (P_1 (n+1) + P_2) 2^{n-1}$$

$$(P_1 n + P_2) 2^n + (P_1 (n+1) + P_2) 2^{n-1} - P_1 2^{n-1} = 3n 2^n$$

$$(P_1 n + P_2) 2^{n-1} [2+1] - P_1 2^{n-1} = 3n 2^n$$

$$3 \cdot 2^{n-1} [P_1 (n) + P_2] - P_1 2^{n-1} = 3n 2^n$$

$$3P_1 n + 3P_2 - P_1 = 6n$$

$$3P_1 = 6$$

$$P_1 = 2$$

$$-3P_2 - P_1 = 0$$

$$P_2 = -2/3$$

$$\Rightarrow a_n = \left( 2n + \frac{-2}{3} \right) 2^n$$

$$Q \quad a_n - 4a_{n-1} + 4a_{n-2} = (n+1) \cdot 2^n$$

$$\beta = 2 \quad \alpha = 1$$

$$a_n - 4a_{n-1} + 4a_{n-2} = 0$$

$$x^n - 4x^{n-1} + 4x^{n-2} = 0$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$x = 2$$

$$x = 2 = \beta$$

$$\Rightarrow m = 2 \quad \because (x-2)^2 \text{ hai}$$

$$a_n = n^2(P_1 n + P_2) 2^n$$

$$a_{n-1} = n^2[P_1(n-1) + P_2] 2^{n-1}$$

$$a_{n-2} = n^2[P_1(n-2) + P_2] 2^{n-2}$$

$$a_n = (P_1 n^3 + P_2 n^2) 2^n$$

$$a_{n-1} = (P_1 n^3 - P_1 n^2 + P_2 n) 2^{n-1}$$

$$a_{n-2} = (P_1 n^3 - 2P_1 n^2 + P_2 n) 2^{n-2}$$

$$\Rightarrow (P_1 n^3 + P_2 n^2) 2^n - 4(P_1 n^3 + P_2 n^2) 2^{n-1} + 4(P_1 n^3 + P_2 n^2) 2^{n-2} + 4P_1 n^2 2^{n-1} - 8P_1 n^2 2^{n-2} = (n+1) 2^n$$

$$\Rightarrow 2^{n-2} (P_1 n^3 + P_2 n^2) [4 - 8 + 4] + 4 \cdot 2 P_1 n^2 2^{n-1} = (n+1) 2^n$$

$$\Rightarrow 4P_1 n^2 2^n = (n+1) 2^n$$

$$4P_1 n^2 - n - 1 = 0$$

$$n(4P_1 n - 1) = 1$$

on solving  $P_1 = \frac{1}{6}$  &  $P_2 = 1$

$$a_n = \left[ \left( \frac{1}{6} \right) n^3 + n^2 \right] 2^n$$



$$Q \quad a_n = a_{n-1} + 7$$

$$a_n - a_{n-1} = 7 \times 1^n \quad \alpha = 0 \quad \beta = 1$$

$$a_n - a_{n-1} = 0$$

$$x^n - x^{n-1} = 0$$

$$x - 1 = 0$$

$$x = 1 = \beta \quad \rightarrow m = 1$$

$$a_n = n P_1$$

$$a_{n-1} = (n-1) P_1$$

$$n P_1 = (n-1) P_1 + 7$$

$$P_1 = 7$$

$$\Rightarrow a_n = 7n$$

Q At person deposits ₹10,000 in bank giving 11% per annum with interest compounded annually. How much will be amount in 30 years.

Let deposited ₹ =  $P_0$  = ₹10,000

$$\text{Also, } P_n = P_{n-1} + \frac{11}{100} P_{n-1}$$

$$\Rightarrow P_n = 1.11 P_{n-1}$$

$$\Rightarrow P_n = (1.11)^2 P_{n-2}$$

$$\Rightarrow P_n = (1.11)^K P_{n-K}$$

$$n-K=0 \quad n=K$$

$$\Rightarrow P_n = (1.11)^n P_0$$

$$\Rightarrow P_n = (1.11)^n \times 10,000$$

$$\Rightarrow P_{30} = (1.11)^{30} \times 10,000$$

Q Suppose that no. of bacteria triples every hour

a) Recurrence relation

b) If 100 bacteria is used at start then how many in 10 hours

$$P_n = 3P_{n-1} = 3^2 \times P_{n-2} = 3^n P_0$$

$$\Rightarrow P_n = 3^n P_0$$

$$\Rightarrow P_n = 3^n (100)$$

$$P_{10} = 3^{10} (100)$$

Q A factory makes cars at an increasing rate. In first month 1 car, 2nd month 2 cars and so on cars in  $n^{\text{th}}$  month. How many cars in 12 months

$$P_n = P_{n-1} + n$$

$$= P_{n-2} + 2n - 1$$

Total cars upto 12 months

$$1 + 2 + \dots + 12$$

$$= \frac{12(13)}{2} = 78 \text{ cars}$$



→ Generating Functions

$$G(z) = a_0 + a_1 z + a_2 z^2 + \dots$$

Q  $a_n = 3a_{n-1} + 2 \quad n \geq 1, a_0 = 0$

Step-1 Multiply eqn with  $z^n$

Step-2 Take  $\sum$  from given limit to  $\infty$

Step-3 Put values in terms of  $G(z)$

Step-4 Find  $G(z)$  in partial fractions

Step-5 Compare with sum of a series

Step-6 Deduce the series

Step-7 Coefficients of terms of  $z$  are summed to get  $a_n$

$$a_n = 3a_{n-1} + 2$$

$$a_n z^n = 3a_{n-1} z^n + 2 z^n$$

$$\sum_{n=1}^{\infty} a_n z^n = 3 \sum_{n=1}^{\infty} a_{n-1} z^n + 2 \sum_{n=1}^{\infty} z^n$$

$$\begin{aligned} & \downarrow \\ & a_1 z + a_2 z^2 + \dots \\ & = G(z) - a_0 \end{aligned}$$

$$\begin{aligned} & \downarrow \\ & a_0 z + a_1 z^2 + \dots \\ & = z G(z) \end{aligned}$$

$$\begin{aligned} & \downarrow \\ & z + z^2 + \dots \\ & = \frac{z}{1-z} \end{aligned}$$

$$\Rightarrow G(z) - a_0 = 3z G(z) + \frac{2z}{1-z}$$

$$\Rightarrow G(z) [1 - 3z] = \frac{2z}{1-z} + a_0$$

$$\Rightarrow G(z) = \frac{2z}{(1-z)(1-3z)} + \frac{a_0}{1-3z}$$

Now  $\frac{2z}{(1-z)(1-3z)} = \frac{A}{1-z} + \frac{B}{1-3z}$

$$2z = A(1-3z) + B(1-z)$$

$$A+B=0 \quad -3A-B=2$$

$$A = -B$$

$$A = -1 \text{ find } B = 1$$

$$G(z) = \frac{-1}{1-z} + \frac{1}{1-3z} \quad (a_0=0)$$

$$\text{series } 1-3z \text{ for } 1-z \text{ and } 1-3z \text{ and } 1-3z$$

$$G(z) = \frac{-1}{1-z} + \frac{1}{1-3z}$$

$$\Rightarrow G(z) = \sum_{n=0}^{\infty} 3^n z^n - \sum_{n=0}^{\infty} 1^n z^n$$

$$\Rightarrow G(z) = \sum_{n=0}^{\infty} 3^n z^n - \sum_{n=0}^{\infty} 1^n z^n$$

$$n=0: 3^0 - 1^0 = 0$$

$$a_n = 3^n - 1^n$$

$$= 3^n - 1$$

$$\frac{z}{1-3z} + \frac{z}{1-z} = \frac{z}{(1-3z)(1-z)}$$

PTO

$$\frac{z}{1-3z} = \frac{z}{(1-3z)(1-z)}$$

$$\frac{z}{1-3z} + \frac{z}{(1-3z)(1-z)}$$



\* Q

$$a_n = a_{n-1} + 2n + 3 \quad n \geq 1 \quad a_0 = 4$$

$$\sum_{n=1}^{\infty} a_n z^n = \sum_{n=1}^{\infty} a_{n-1} z^n + 2 \sum_{n=1}^{\infty} n z^n + 3 \sum_{n=1}^{\infty} z^n$$

$$a_1 z + a_2 z^2 + \dots$$

$$G(z) - a_0$$

$$a_0 z + a_1 z^2 + \dots$$

$$z G(z)$$

$$z + 2z^2 + \dots$$

$$AGP = z$$

$$(1-z)^2$$

$$z + z^2 + \dots$$

$$= \frac{z}{1-z}$$

$$\Rightarrow G(z) - a_0 = z G(z) + \frac{2z}{(1-z)^2} + \frac{3z}{(1-z)}$$

$$\Rightarrow G(z) = \frac{2z}{(1-z)^3} + \frac{3z}{(1-z)^2} + \frac{4}{(1-z)}$$

\* Solution for  $\frac{1}{(1-z)^m}$

$$\frac{1}{(1-z)^m} = \sum_{k=0}^{\infty} \binom{m+k-1}{k} z^k = \sum_{k=0}^{\infty} \binom{m+k-1}{m-1} z^k$$

$$\Rightarrow \frac{1}{(1-z)^3} = \sum_{k=0}^{\infty} \binom{k+2}{2} z^k = \sum_{k=0}^{\infty} \frac{(k+2)!}{2! k!} z^k$$

$$= \sum_{k=0}^{\infty} \left[ \frac{k^2 + 3k + 2}{2} \right] z^k$$

$$\Rightarrow \frac{1}{(1-z)^3} = \frac{z}{(1-z)^3} + z \sum_{k=0}^{\infty} \binom{k+2}{2} z^k$$

$$= \sum_{k=0}^{\infty} \frac{k+2}{2} c_2 z^{k+1}$$

$$= \sum_{k=0}^{\infty} \left[ \frac{k^2 + 3k + 2}{2} \right] z^{k+1}$$

Replace  $k$  with  $k+1$

$$\Rightarrow \sum_{k=1}^{\infty} \left[ \frac{k^2 + 1 - 2k + 3k - 3 + 2}{2} \right] z^k$$

$$= \sum_{k=1}^{\infty} \left[ \frac{k^2 + k}{2} \right] z^k + 0$$

$k=0, \left( \frac{k^2 + k}{2} \right) z^k = 0$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{k^2 + k}{2} z^k$$

$$\Rightarrow G(z) = 2 \sum_{k=0}^{\infty} \frac{k^2 + k}{2} z^k + 3 \sum_{k=0}^{\infty} k z^k + 4 \sum_{k=0}^{\infty} z^k$$

$$\Rightarrow a_k = k^2 + k + 3k + 4$$

$$\Rightarrow a_n = n^2 + 4n + 4$$

$$Q \quad a_n = n+1$$

$$G(z) = a_0 + a_1 z + a_2 z^2 + \dots$$

$$G(z) = 1 + 2z + 3z^2 + \dots$$

$$z G(z) = z + 2z^2 + 3z^3 + \dots$$

$$(1-z) G(z) = 1 + z + z^2 + \dots$$

$$G(z) = \frac{1}{(1-z)^2}$$

$$= \sum_{n=0}^{\infty} 2^n z^n + \sum_{n=0}^{\infty} 3^n z^n$$



$$\Rightarrow a_n = 2^n + 3^n$$

$$Q \quad a_{n+2} - 5a_{n+1} + 6a_n = 2 \quad a_0 = 1 \quad a_1 = 2$$

$$a_n - 5a_{n-1} + 6a_{n-2} = 2 \quad n \geq 2$$

$$\sum_{n=2}^{\infty} a_n z^n - 5 \sum_{n=2}^{\infty} a_{n-1} z^n + 6 \sum_{n=2}^{\infty} a_{n-2} z^n = 2 \sum_{n=2}^{\infty} z^n$$

$$a_2 z^2 + a_3 z^3 + \dots \quad a_1 z^2 + a_2 z^3 + \dots \quad a_0 z^2 + a_1 z^3 + \dots$$

$$z G(z) - a_0 z \quad z^2 G(z)$$

$$G(z) - a_0 - a_1 z - (5z G(z) - 5a_0 z) + 6z^2 G(z) = \frac{2z^2}{1-z}$$

$$G(z) - 1 - 2z - 5z G(z) + 5z + 6z^2 G(z) = \frac{2z^2}{1-z}$$

$$\Rightarrow G(z) [1 + 5z + 6z^2] = \frac{2z^2}{1-z} + 1 - 3z$$

$$\Rightarrow G(z) \cdot (3z-1)(2z-1) = \frac{2z^2}{1-z} + 1 - 3z$$

$$\Rightarrow G(z) = \frac{2z^2}{(1-z)(2z-1)(3z-1)} - \frac{1}{2z-1}$$

$$\Rightarrow G(z) = \frac{2z^2}{(1-z)(1-2z)(1-3z)} + \frac{1}{1-2z}$$

$$= \frac{2z^2 + 1 - 3z - z + 3z^2}{(1-z)(1-2z)(1-3z)}$$

$$= \frac{5z^2 - 4z + 1}{(1-z)(1-2z)(1-3z)}$$

$$= \frac{A}{1-z} + \frac{B}{1-2z} + \frac{C}{1-3z}$$

$$\begin{aligned} \Rightarrow 5z^2 - 4z + 1 &= A(1-2z)(1-3z) + B(1-z)(1-3z) + C(1-z)(1-2z) \\ &= A(1-3z-2z+6z^2) + B(1-3z-z+3z^2) + C(1-2z-z+2z^2) \\ &= A(6z^2-5z+1) + B(3z^2-4z+1) + C(2z^2-3z+1) \\ &= z^2(6A+3B+2C) + z(-5A-4B-3C) + (A+B+C) \end{aligned}$$

$$6A + 3B + 2C = 5$$

$$5A + 4B + 3C = 4$$

$$A + B + C = 1$$

$$5(1-B-C) + 4B + 3C = 4$$

$$5 - 5B - 5C + 4B + 3C = 4$$

$$-5B - 2C = -1 \Rightarrow B + 2C = 1$$

$$B + 2C = 1$$

$$A = 1 - B - C$$

$$B = 1 - 2C$$

$$A = 1 - 1 + 2C - C = C$$

$$6C + 3 - 6C + 2C = 5$$

$$C = 1$$

$$A = 1$$

$$B = -1$$



$\Sigma \rightarrow \text{simult}$

1/1

$$\Rightarrow G(z) = \frac{1}{1-z} - \frac{1}{1-2z} + \frac{1}{1-3z}$$

$$= \sum_{n=0}^{\infty} z^n - \sum_{n=0}^{\infty} (2z)^n + \sum_{n=0}^{\infty} (3z)^n$$

$$\Rightarrow a_n = 1^n - 2^n + 3^n$$

$$\Rightarrow a_n = 1 - 2^n + 3^n$$

prüfen ob es für alle  $n$  gilt

prüfen ob es für alle  $n$  gilt

prüfen ob es für alle  $n$  gilt



prüfen ob es für alle  $n$  gilt

$\Sigma = (i)$  vergrößern

(1. Abkürzung)  $1 = (1)$  vergrößern  
(2. Abkürzung)  $1 = (1)$  vergrößern